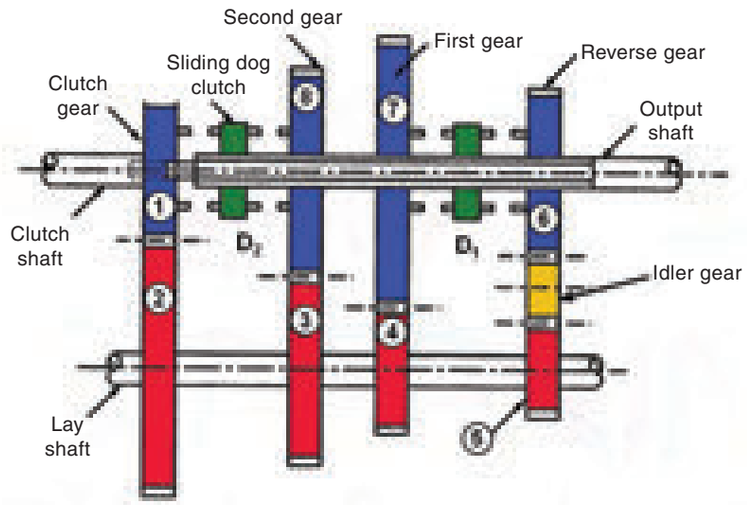


Figure 3.5.2 (a) Gear arrangements in the constant mesh gear box

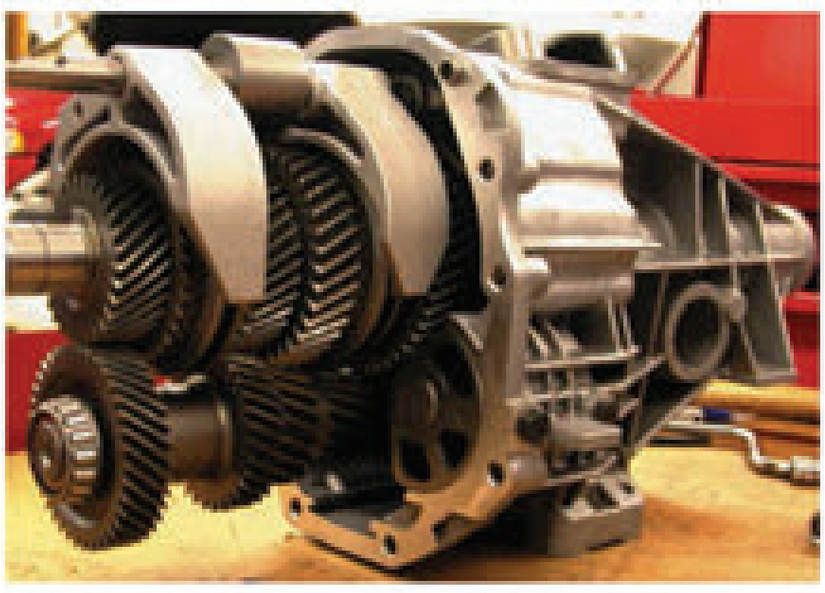


Figure 3.5.2 (b) Constant mesh gear box

are allowed to move on the slots made on the main output shaft. All the shafts in the gear box are supported by the bearings in the housing.

Working principle

Neutral position

During neutral condition both the dog clutches (D_1 and D_2) are not in mesh with any gears as shown in Figure. All gears in the shafts 1,2 & 3,4 & 5,6 & 7,8,9 rotate. However the main output shaft does not rotate as the dog clutches are not in engagement with any of the gears. Hence rotational power is not transmitted.

First gear

By actuating the gear shift lever, the dog clutch D_1 is moved towards left

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